E Notions of Information Theory

This chapter introduces some basic notions of information theory useful for the presentation of several learning algorithms and their properties. The definitions and theorems are given in the case of discrete random variables or distributions, but they can be straightforwardly extended to the continuous case.

We start with the notion of *entropy*, which can be viewed as a measure of the uncertainty of a random variable.

E.1 Entropy

Definition E.1 (Entropy) *The* entropy *of a discrete random variable X with probability mass function $p(x) = \mathbb{P}[X = x]$ is denoted by $\mathsf{H}(X)$ and defined by*

$$\mathsf{H}(X) = -\mathbb{E}\left[\log(p(X))\right] = -\sum_{x \in \mathcal{X}} p(x) \log(p(x)). \tag{E.1}$$

We define by the same expression the entropy of a distribution *p and abusively denote that by $\mathsf{H}(p)$.*

The base of the logarithm is not critical in this definition since it only affects the value by a multiplicative constant. Thus, unless otherwise specified, we will consider the natural logarithm (base e). If we use base 2, then $-\log_2(p(x))$ is the number of bits needed to represent $p(x)$. Thus, by definition, the entropy of X can be viewed as the average number of bits (or amount of information) needed for the description of the random variable X. By the same property, the entropy is always non-negative:

$$\mathsf{H}(X) \geq 0. \tag{E.2}$$

As an example, the entropy of a biased coin X_p taking value 1 with probability p and 0 with probability $1 - p$ is given by

$$\mathsf{H}(X_p) = -p \log p - (1 - p) \log(1 - p). \tag{E.3}$$

The corresponding function of p is often referred to as the *binary entropy function*. Figure E.1 shows a plot of that function when using base 2 for the logarithm. As can be seen from the figure, the entropy is a concave function.[26] It reaches its maximum at $p = \frac{1}{2}$, which corresponds to the most uncertain case, and its minima at $p = 0$ or $p = 1$ which correspond to the fully certain cases. More generally, assume that the input space $\mathcal{X}$ has a finite cardinality $N \geq 1$. Then, by Jensen's

[26] We will see later that the entropy function is always concave.

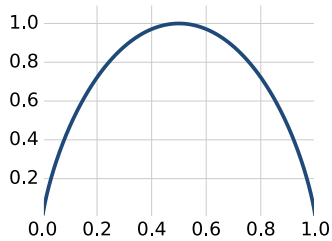

Figure E.1
A plot of the binary entropy as a function of the bias p.

inequality, in view of the concavity of logarithm, the following inequality holds:

$$\mathsf{H}(X) = \mathbb{E}\left[\log \frac{1}{p(X)}\right] \leq \log \mathbb{E}\left[\frac{1}{p(X)}\right] = \log\left(\sum_{x\in\mathcal{X}} \frac{p(x)}{p(x)}\right) = \log N. \tag{E.4}$$

Thus, more generally, the maximum value of the entropy is $\log N$, that is the entropy of the uniform distribution.

The entropy is a lower bound on lossless data compression and is therefore a critical quantity to consider in information theory. It is also closely related to the notions of entropy in thermodynamics and quantum physics.

E.2 Relative entropy

Here, we introduce a measure of divergence between two distributions p and q, *relative entropy*, which is related to the notion of entropy. The following is its definition in the discrete case.

Definition E.2 (Relative entropy) *The* relative entropy *(or Kullback-Leibler divergence) of two distributions p and q is denoted by $\mathsf{D}(p\|q)$ and defined by*

$$\mathsf{D}(p\|q) = \mathop{\mathbb{E}}_{p}\left[\log \frac{p(X)}{q(X)}\right] = \sum_{x\in\mathcal{X}} p(x) \log\left[\frac{p(x)}{q(x)}\right], \tag{E.5}$$

with the conventions $0\log 0 = 0$, $0\log\frac{0}{0} = 0$, and $a\log\frac{a}{0} = +\infty$ for $a > 0$.

Note that, in view of these conventions, whenever $q(x) = 0$ for some x in the support of p ($p(x) > 0$), the relative entropy is infinite: $\mathsf{D}(p\|q) = \infty$. Thus, the relative entropy does not provide an informative measure of the divergence of p and q in such cases.

As for the entropy, the base of the logarithm is not critical in the definition of the relative entropy and we will consider the natural logarithm unless otherwise specified. If we use base 2, the relative entropy can be interpreted in terms of coding length. Ideally, one could design for p an optimal code with average length the entropy $\mathsf{H}(p)$. The relative entropy is the average number of additional bits needed to encode p when using an optimal code for q instead of one for p since it can be expressed as the difference $\mathsf{D}(p\|q) = \mathbb{E}_p[\log\frac{1}{q(X)}] - \mathsf{H}(p)$, which, as shown by the following proposition, is always non-negative.

Proposition E.3 (Non-negativity of relative entropy) *For any two distributions p and q, the following inequality holds:*

$$\mathsf{D}(p\|q) \geq 0. \tag{E.6}$$

Furthermore, $\mathsf{D}(p\|q) = 0$ iff $p = q$.

Proof: By the concavity of logarithm and Jensen's inequality, the following holds:

$$-\mathsf{D}(p\|q) = \sum_{x\colon p(x)>0} p(x)\log\left(\frac{q(x)}{p(x)}\right) \le \log\left(\sum_{x\colon p(x)>0} p(x)\frac{q(x)}{p(x)}\right)$$
$$= \log\left(\sum_{x\colon p(x)>0} q(x)\right) \le \log(1) = 0.$$

Thus, the relative entropy is always non-negative for all distributions p and q. The equality $\mathsf{D}(p\|q) = 0$ can hold only if both of the inequalities above are equalities. The last one implies that $\sum_{x\colon p(x)>0} q(x) = 1$. Since the log function is strictly concave, the first inequality can be an equality only if $\frac{q(x)}{p(x)}$ is some constant α over $\{x\colon p(x) > 0\}$. Since $p(x)$ sums to one over that set, we must have $\sum_{x\colon p(x)>0} q(x) = \alpha$. Thus, $\alpha = 1$, which implies $q(x) = p(x)$ for all $x \in \{x\colon p(x) > 0\}$ and thus for all x. Finally, by definition, for any distribution p, $\mathsf{D}(p\|p) = 0$, which completes the proof. □

The relative entropy is not a distance. It is asymmetric: in general, $\mathsf{D}(p\|q) \neq \mathsf{D}(q\|p)$ for two distributions p and q. Furthermore, in general, the relative entropy does not verify the triangle inequality.

Corollary E.4 (Log-sum inequality) *For any set of non-negative real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$, the following inequality holds:*

$$\sum_{i=1}^{n} a_i \log\left(\frac{a_i}{b_i}\right) \ge \left(\sum_{i=1}^{n} a_i\right)\log\left(\frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i}\right), \tag{E.7}$$

with the conventions $0\log 0 = 0$, $0\log\frac{0}{0} = 0$, and $a\log\frac{a}{0} = +\infty$ for $a > 0$.

Furthermore, equality holds in (E.7) iff $\frac{a_i}{b_i}$ is a constant (does not depend on i).

Proof: With the conventions adopted, it is clear that the equality holds if $\sum_{i=1}^{n} a_i = 0$, that is $a_i = 0$ for all $i \in [n]$, or $\sum_{i=1}^{n} b_i = 0$, that is $b_i = 0$ for all $i \in [n]$. Thus, we can assume that $\sum_{i=1}^{n} a_i \neq 0$ and $\sum_{i=1}^{n} b_i \neq 0$. Since the inequality is invariant by scaling of the a_is or b_is, we can multiply them by positive constants such that $\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} b_i = 1$. The inequality then coincides with the non-negativity of the relative entropy of the distributions thereby defined by a_is and b_is and the result holds by Proposition E.3. □

Corollary E.5 (Joint convexity of relative entropy) *The relative entropy function $(p, q) \mapsto \mathsf{D}(p\|q)$ is convex.*

Proof: For any $\alpha \in [0, 1]$ and any four probability distributions p_1, p_2, q_1, q_2, by the Log-sum inequality (Corollary E.4), the following holds for any fixed x:

$$(\alpha p_1(x) + (1-\alpha)p_2(x))\log\left[\frac{\alpha p_1(x) + (1-\alpha)p_2(x)}{\alpha q_1(x) + (1-\alpha)q_2(x)}\right]$$
$$\le \alpha p_1(x)\log\left[\frac{\alpha p_1(x)}{\alpha q_1(x)}\right] + (1-\alpha)p_2(x)\log\left[\frac{(1-\alpha)p_1(x)}{(1-\alpha)q_2(x)}\right]. \tag{E.8}$$

Summing up these inequalities over all x yields:

$$\mathsf{D}\Big(\alpha p_1 + (1-\alpha)p_2 \| \alpha q_1 + (1-\alpha)q_2\Big) \le \alpha\mathsf{D}(p_1\|q_1) + (1-\alpha)\mathsf{D}(p_2\|q_2), \tag{E.9}$$

which concludes the proof. □

Corollary E.6 (Concavity of the entropy) *The entropy function $p \mapsto \mathsf{H}(p)$ is concave.*

Proof: Observe that for any fixed distribution p_0 over $\mathcal{X}$, by definition of the relative entropy, we can write

$$\mathsf{D}(p\|p_0) = \sum_{x\in\mathcal{X}} p(x)\log(p(x)) - \sum_{x\in\mathcal{X}} p(x)\log(p_0(x)). \tag{E.10}$$

Thus, $\mathsf{H}(p) = -\mathsf{D}(p\|p_0) - \sum_{x\in\mathcal{X}} p(x)\log(p_0(x))$. By Corollary E.5, the first term is a concave function of p. The second term is linear in p and therefore is concave. Thus, H is concave as a sum of two concave functions. □

Proposition E.7 (Pinsker's inequality) *For any two distributions p and q, the following inequality holds:*

$$\mathsf{D}(p\|q) \geq \frac{1}{2}\|p-q\|_1^2. \tag{E.11}$$

Proof: We first show that the inequality holds for distributions over a set $\mathcal{A} = \{a_0, a_1\}$ of cardinality 2. Let $p_0 = p(a_0)$ and $q_0 = q(a_0)$. Fix $p_0 \in [0,1]$ and consider the function $f\colon q_0 \mapsto f(q_0)$ defined by

$$f(q_0) = p_0 \log\frac{p_0}{q_0} + (1-p_0)\log\frac{1-p_0}{1-q_0} - 2(p_0-q_0)^2. \tag{E.12}$$

Observe that $f(p_0) = 0$ and that for $q_0 \in (0,1)$,

$$f'(q_0) = -\frac{p_0}{q_0} + \frac{1-p_0}{1-q_0} + 4(p_0-q_0) = (q_0-p_0)\left[\frac{1}{(1-q_0)q_0} - 4\right]. \tag{E.13}$$

Since $(1-q_0)q_0 \leq \frac{1}{4}$, $[\frac{1}{(1-q_0)q_0} - 4]$ is non-negative. Thus, $f'(q_0) \leq 0$ for $q_0 \leq p_0$ and $f'(q_0) \geq 0$ for $q_0 \geq p_0$. Thus, f reaches its minimum at $q_0 = p_0$, which implies $f(q_0) \geq f(p_0) = 0$ for all q_0. Since $f(q_0)$ can be expressed as follows:

$$\begin{aligned} f(q_0) &= \mathsf{D}(p\|q) - 2(p_0-q_0)^2 && \text{(E.14)}\\ &= \mathsf{D}(p\|q) - \frac{1}{2}\Big[|p_0-q_0| + |(1-p_0)-(1-q_0)|\Big]^2 && \text{(E.15)}\\ &= \mathsf{D}(p\|q) - \frac{1}{2}\|p-q\|_1^2 \geq 0, && \text{(E.16)} \end{aligned}$$

this proves the inequality for a set $\mathcal{A} = \{a_0, a_1\}$ of cardinality 2.

Now, consider the distributions p' and q' defined over $\mathcal{A} = \{a_0, a_1\}$ with $p'(a_0) = \sum_{x\in a_0} p(x)$, and $q'(a_0) = \sum_{x\in a_0} q(x)$ where $a_0 = \{x \in \mathcal{X}\colon p(x) \geq q(x)\}$ and $a_1 = \{x \in \mathcal{X}\colon p(x) < q(x)\}$. By the Log-sum inequality (Corollary E.4),

$$\begin{aligned} \mathsf{D}(p\|q) &= \sum_{x\in a_0} p(x)\log\left[\frac{p(x)}{q(x)}\right] + \sum_{x\in a_1} p(x)\log\left[\frac{p(x)}{q(x)}\right] && \text{(E.17)}\\ &\geq p(a_0)\log\left[\frac{p(a_0)}{q(a_0)}\right] + p(a_1)\log\left[\frac{p(a_1)}{q(a_1)}\right] && \text{(E.18)}\\ &= \mathsf{D}(p'\|q'). && \text{(E.19)} \end{aligned}$$

Combining this inequality with the observation that

$$\begin{aligned} \|p'-q'\|_1 &= (p(a_0)-q(a_0)) - (p(a_1)-q(a_1)) && \text{(E.20)}\\ &= \sum_{x\in a_0}(p(x)-q(x)) - \sum_{x\in a_1}(p(x)-q(x)) && \text{(E.21)}\\ &= \sum_{x\in\mathcal{X}}|p(x)-q(x)| && \text{(E.22)}\\ &= \|p-q\|_1, && \text{(E.23)} \end{aligned}$$

shows that $\mathsf{D}(p\|q) \geq \mathsf{D}(p'\|q') \geq \frac{1}{2}\|p-q\|_1^2$ and concludes the proof. □

Definition E.8 (Conditional relative entropy) *Let p and q be two probability distributions defined over $\mathcal{X}\times\mathcal{Y}$ and r a distribution over $\mathcal{X}$. Then, the* conditional relative entropy *of p and q with respect to the marginal r is defined as the expectation of the relative entropy of $p(\cdot|X)$ and $q(\cdot|X)$ with respect to r:*

$$\mathop{\mathbb{E}}_{X\sim r}\Big[\mathsf{D}\big(p(\cdot|X)\|q(\cdot|X)\big)\Big] = \sum_{x\in\mathcal{X}} r(x)\sum_{y\in\mathcal{Y}} p(y|x)\log\frac{p(y|x)}{q(y|x)} = \mathsf{D}(\widetilde{p}\|\widetilde{q}), \tag{E.24}$$

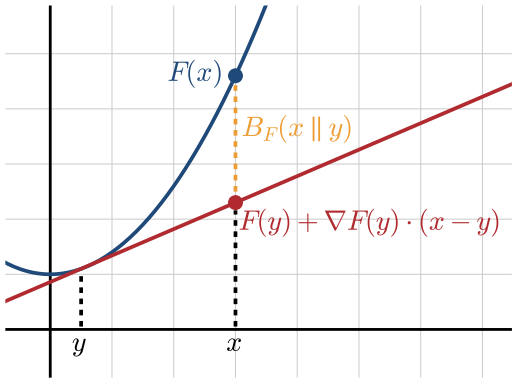

Figure E.2
Illustration of the quantity measured by the Bregman divergence defined based on a convex and differentiable function F. The divergence measures the distance between $F(x)$ and the hyperplane tangent to the curve at point y.

where $\widetilde{p}(x, y) = r(x)p(y|x)$ *and* $\widetilde{q}(x, y) = r(x)q(y|x)$*, with the conventions* $0 \log 0 = 0$*,* $0 \log \frac{0}{0} = 0$*, and* $a \log \frac{a}{0} = +\infty$ *for* $a > 0$*.*

E.3 Mutual information

Definition E.9 (Mutual information) *Let* X *and* Y *be two random variables with joint probability distribution function* $p(\cdot, \cdot)$ *and marginal probability distribution functions* $p(x)$ *and* $p(y)$*. Then, the* mutual information *of* X *and* Y *is denoted by* $I(X, Y)$ *and defined as follows:*

$$I(X, Y) = \mathsf{D}(p(x, y) \| p(x)p(y)) \tag{E.25}$$

$$= \mathop{\mathbb{E}}_{p(x,y)} \left[\log \frac{p(X, Y)}{p(X)p(Y)} \right] = \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x, y) \log \left[\frac{p(x, y)}{p(x)p(y)} \right], \tag{E.26}$$

with the conventions $0 \log 0 = 0$*,* $0 \log \frac{0}{0} = 0$*, and* $a \log \frac{a}{0} = +\infty$ *for* $a > 0$*.*

When the random variables X and Y are independent, their joint distributions is the product of the marginals $p(x)$ and $p(y)$. Thus, the mutual information is a measure of the closeness of the joint distribution $p(x, y)$ to its value when X and Y are independent, where closeness is measured via the relative entropy divergence. As such, it can be viewed as a measure of the amount of information that each random variable can provide about the other. Note that by Proposition E.3, the equality $I(X, Y) = 0$ holds iff $p(x, y) = p(x)p(y)$ for all x, y, that is iff X and Y are independent.

E.4 Bregman divergences

Here we introduce the so-called *unnormalized relative entropy* $\widetilde{\mathsf{D}}$ defined for all non-negative functions p, q in $\mathbb{R}^{\mathcal{X}}$ by

$$\widetilde{\mathsf{D}}(p \| q) = \sum_{x \in \mathcal{X}} p(x) \log \left[\frac{p(x)}{q(x)} \right] + \big(q(x) - p(x)\big), \tag{E.27}$$

Table E.1
Examples of Bregman divergences and corresponding convex functions.

	$\mathsf{B}_F(x \,\|\, y)$	$F(x)$
Squared L_2-distance	$\|x-y\|^2$	$\|x\|^2$
Mahalanobis distance	$(x-y)^\top Q(x-y)$	$x^\top Qx$
Unnormalized relative entropy	$\widetilde{\mathsf{D}}(x \,\|\, y)$	$\sum_{i\in I} x(i)\log(x(i)) - x(i)$

with the conventions $0\log 0 = 0$, $0\log\frac{0}{0} = 0$, and $a\log\frac{a}{0} = +\infty$ for $a > 0$. The relative entropy coincides with the unnormalized relative entropy when restricted to $\Delta\times\Delta$, where Δ is the family of distributions defined over $\mathcal{X}$. The relative entropy inherits several properties of the unnormalized relative entropy, in particular, it can be shown that $\widetilde{\mathsf{D}}(p\|q) \geq 0$. Many of these properties are in fact shared by a broader family of divergences known as *Bregman divergences*.

Definition E.10 (Bregman divergences) *Let F be a convex and differentiable function defined over a convex (open) set $\mathcal{C}$ in a Hilbert space $\mathbb{H}$. Then, the* Bregman divergence B_F *associated to F is defined for all $x, y \in \mathcal{C}$ by*

$$\mathsf{B}_F(x\|y) = F(x) - F(y) - \langle \nabla F(y), x-y\rangle\,. \tag{E.28}$$

Thus, $\mathsf{B}_F(x \,\|\, y)$ measures the difference of $F(x)$ and its linear approximation. Figure E.2 illustrates this definition. Table E.1 provides several examples of Bregman divergences along with their corresponding convex functions $F(x)$. Note that, although the unnormalized relative entropy is a Bregman divergence, the relative entropy is not a Bregman divergence since it is defined over the simplex which is not an open set and has an empty interior.

The following proposition presents several general properties of Bregman divergences.

Proposition E.11 *Let F be a convex and differentiable function defined over a convex set $\mathcal{C}$ in a Hilbert space $\mathbb{H}$. Then, the following properties hold:*

1. $\forall x, y \in \mathcal{C},\ \mathsf{B}_F(x \,\|\, y) \geq 0$.

2. $\forall x, y, z \in \mathcal{C}, \langle \nabla F(x) - \nabla F(y), x - z\rangle = \mathsf{B}_F(x \,\|\, y) + \mathsf{B}_F(z \,\|\, x) - \mathsf{B}_F(z \,\|\, y)$.

3. *B_F is convex in its first argument. If additionally F is strictly convex, then B_F is strictly convex in its first argument.*

4. *Linearity: let G be a convex and differentiable function over $\mathcal{C}$, then, for any $\alpha, \beta \in \mathbb{R}$, $\mathsf{B}_{\alpha F+\beta G} = \alpha\mathsf{B}_F + \beta\mathsf{B}_G$.*

   For the following properties, we will assume additionally that F is strictly convex.

5. *Projection: for any $y \in \mathcal{C}$ and any closed convex set $\mathcal{K} \subseteq \mathcal{C}$, the B_F-projection of y over K, $P_{\mathcal{K}}(y) = \operatorname{argmin}_{x\in\mathcal{K}} \mathsf{B}_F(x \,\|\, y)$, is unique.*

6. *Pythagorean theorem: for $y \in \mathcal{C}$ and any closed convex set $\mathcal{K} \subseteq \mathcal{C}$, the following holds for all $x \in \mathcal{K}$:*
$$\mathsf{B}_F(x \,\|\, y) \geq \mathsf{B}_F(x \,\|\, P_{\mathcal{K}}(y)) + \mathsf{B}_F(P_{\mathcal{K}}(y) \,\|\, y). \tag{E.29}$$

7. *Conjugate divergence: assume that F is closed proper strictly convex, and that the norm of its gradient tends to infinity near the boundary of $\mathcal{C}$: $\lim_{x\to\partial\mathcal{C}} \|\nabla F(x)\| = +\infty$. The pair $(\mathcal{C}, F)$ is then said to be a* convex function of Legendre type. *Then, the conjugate of F, F^*, is differentiable and the following holds for all $x, y \in \mathcal{C}$:*
$$\mathsf{B}_F(x \,\|\, y) = \mathsf{B}_{F^*}(\nabla F(y) \,\|\, \nabla F(x)). \tag{E.30}$$

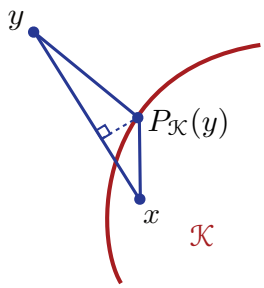

Figure E.3
A depiction of the Pythagorean theorem stated in proposition E.11, where the squared length of each line illustrates the magnitude of the Bregman divergence between the points it connects.

Proof: Property (1) holds by convexity of the function F (the graph of F is above its tangent, see equation (B.3)).

Property (2) follows directly from the definition of the Bregman divergence:

$$\begin{aligned}
&\mathsf{B}_F(x \,\|\, y) + \mathsf{B}_F(z \,\|\, x) - \mathsf{B}_F(z \,\|\, y) \\
&\quad = -\langle \nabla F(y), x - y\rangle - \langle \nabla F(x), z - x\rangle + \langle \nabla F(y), z - y\rangle \\
&\quad = \langle \nabla F(x) - \nabla F(y), x - z\rangle \ .
\end{aligned}$$

Property (3) holds since $x \mapsto F(x) - F(y) - \langle \nabla F(y), x - y\rangle$ is convex as a sum of the convex function $x \mapsto F(x)$ and the affine and thus convex function $x \mapsto -F(y) - \langle \nabla F(y), x - y\rangle$. Similarly, B_F is strictly convex with respect to its first argument if F is strictly convex, as a sum of a strictly convex function and an affine function.

Property (4) follows from a series of equalities:

$$\begin{aligned}
\mathsf{B}_{\alpha F + \beta G} &= \alpha F(x) + \beta G(x) - \alpha F(y) - \beta G(y) - \big\langle \nabla\big(\alpha F(y) + \beta G(y)\big), x - y\big\rangle \\
&= \alpha\big(F(x) - F(y) - \langle \nabla F(y), x - y\rangle\big) + \beta\big(G(x) - G(y) - \langle \nabla G(y), x - y\rangle\big) \\
&= \alpha \mathsf{B}_F + \beta \mathsf{B}_G,
\end{aligned}$$

where we have used the fact that both the gradient and inner-product are linear functions.

Property (5) holds since, by Property (3), $\min_{x \in \mathcal{K}} \mathsf{B}_F(x \,\|\, y)$ is a convex optimization problem with a strictly convex objective function.

For property (6), fix $y \in \mathcal{C}$ and let J be the function defined for all $\alpha \in [0, 1]$ by

$$J(\alpha) = \mathsf{B}_F(\alpha x + (1 - \alpha) P_{\mathcal{K}}(y) \,\|\, y).$$

Since $\mathcal{C}$ is convex, for any $\alpha \in [0, 1]$, $\alpha x + (1 - \alpha) P_{\mathcal{K}}(y)$ is in $\mathcal{C}$. F is differentiable over $\mathcal{C}$ therefore J is also differentiable as a composition of F with $\alpha \mapsto \alpha x + (1 - \alpha) P_{\mathcal{K}}(y)$. By definition of $P_{\mathcal{K}}(y)$, for any $\alpha \in (0, 1]$,

$$\frac{J(\alpha) - J(0)}{\alpha} = \frac{\mathsf{B}_F(\alpha x + (1 - \alpha) P_{\mathcal{K}}(y) \,\|\, y) - \mathsf{B}_F(P_{\mathcal{K}}(y) \,\|\, y)}{\alpha} \geq 0. \tag{E.31}$$

This implies that $J'(0) \geq 0$. From the following expression of $J(\alpha)$:

$$J(\alpha) = F(\alpha x + (1 - \alpha) P_{\mathcal{K}}(y)) - F(y) - \langle \nabla F(y), \alpha x + (1 - \alpha) P_{\mathcal{K}}(y) - y\rangle \,, \tag{E.32}$$

we can compute its derivative at 0:

$$\begin{aligned}
J'(0) &= \langle x - P_{\mathcal{K}}(y), \nabla F(P_{\mathcal{K}}(y))\rangle - \langle \nabla F(y), x - P_{\mathcal{K}}(y)\rangle \\
&= -\mathsf{B}_F(x \,\|\, P_{\mathcal{K}}(y)) + F(x) - F(P_{\mathcal{K}}(y)) - \langle \nabla F(y), x - P_{\mathcal{K}}(y)\rangle \\
&= -\mathsf{B}_F(x \,\|\, P_{\mathcal{K}}(y)) + F(x) - F(P_{\mathcal{K}}(y)) - \langle \nabla F(y), x - y\rangle - \langle \nabla F(y), y - P_{\mathcal{K}}(y)\rangle \\
&= -\mathsf{B}_F(x \,\|\, P_{\mathcal{K}}(y)) + \mathsf{B}_F(x \,\|\, y) + F(y) - F(P_{\mathcal{K}}(y)) - \langle \nabla F(y), y - P_{\mathcal{K}}(y)\rangle \\
&= -\mathsf{B}_F(x \,\|\, P_{\mathcal{K}}(y)) + \mathsf{B}_F(x \,\|\, y) - \mathsf{B}_F(P_{\mathcal{K}}(y) \,\|\, y) \geq 0,
\end{aligned}$$

which concludes the proof of Property (6).

For property (7), note that, by definition, for any y, F^* is defined by

$$F^*(y) = \sup_{x \in \mathcal{C}} \big\{ \langle x, y \rangle - F(x) \big\}. \tag{E.33}$$

F^* is convex and admits a sub-differential at any y. By the strict convexity of F, the function $x \mapsto \langle x, y \rangle - F(x)$ is strictly concave and differentiable over $\mathcal{C}$ and the norm of its gradient, $y - \nabla F(x)$, tends to infinity near the boundary of $\mathcal{C}$ (by the corresponding property assumed for F). Thus, its supremum is reached at a unique point $x_y \in \mathcal{C}$ where its gradient is zero, that is at x_y with $\nabla F(x_y) = y$. This implies that for any y, $\partial F^*(y)$, the subdifferential of F^*, is reduced to a singleton. Thus, F^* is differentiable and its gradient at y is $\nabla F^*(y) = x_y = \nabla^{-1} F(y)$. Since F^* is convex and differentiable, its Bregman divergence is well defined. Furthermore, $F^*(y) = \big\langle \nabla F^{-1}(y), y \big\rangle - F(\nabla F^{-1}(y))$ since $x_y = \nabla^{-1} F(y)$. For any $x, y \in \mathcal{C}$, using the definition of B_{F^*} and the expression of $\nabla F^*(y)$ and $F^*(y)$ we can write

$$\begin{aligned}
&\mathsf{B}_{F^*}(\nabla F(y) \,\|\, \nabla F(x)) \\
&= F^*(\nabla F(y)) - F^*(\nabla F(x)) - \big\langle \nabla^{-1} F(\nabla F(x)), \nabla F(y) - \nabla F(x) \big\rangle \\
&= F^*(\nabla F(y)) - F^*(\nabla F(x)) - \langle x, \nabla F(y) - \nabla F(x) \rangle \\
&= \big\langle \nabla^{-1} F(\nabla F(y)), \nabla F(y) \big\rangle - F(\nabla^{-1} F(\nabla F(y))) \\
&\qquad - \big\langle \nabla^{-1} F(\nabla F(x)), \nabla F(x) \big\rangle + F(\nabla^{-1} F(\nabla F(x))) - \langle x, \nabla F(y) - \nabla F(x) \rangle \\
&= \langle y, \nabla F(y) \rangle - F(y) - \langle x, \nabla F(x) \rangle + F(x) - \langle x, \nabla F(y) - \nabla F(x) \rangle \\
&= \langle y, \nabla F(y) \rangle - F(y) + F(x) - \langle x, \nabla F(y) \rangle \\
&= F(x) - F(y) - \langle x - y, \nabla F(y) \rangle = \mathsf{B}_F(x \,\|\, y),
\end{aligned}$$

which completes the proof. □

Notice that while the unnormalized relative entropy (and thus the relative entropy) are convex functions of the pair of their arguments, this in general does not hold for all Bregman divergences, only convexity with respect to the first argument is guaranteed.

The notion of Bregman divergence can be extended to the case of non-differentiable functions (see section 14.3).

E.5 Chapter notes

The notion of entropy presented in this chapter is due to Shannon [1948] who, more generally, within the same article, set the foundation of information theory. More general definitions of entropy (*Rényi entropy*) and relative entropy (*Rényi divergence*) were later introduced by Rényi [1961]. The Kullback-Leibler divergence was introduced in [Kullback and Leibler, 1951].

Pinsker's inequality is due to Pinsker [1964]. Finer inequalities relating the relative entropy and the L_1-norm were later given by Csiszár [1967] and Kullback [1967]. See [Reid and Williamson, 2009] for a generalization of such inequalities to the case of f-divergences. The notion of Bregman divergence is due to Bregman [1967].

For a more extensive material on information theory, we strongly recommend the book of Cover and Thomas [2006].

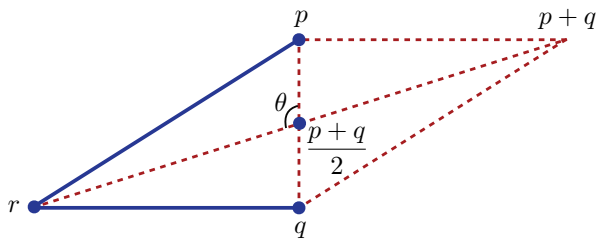

Figure E.4
An illustration of the parallelogram identity.

E.6 Exercises

E.1 Parallelogram identity. Prove the following *parallelogram identity* for any three distributions p, q, and r on $\mathcal{X}$:

$$\mathsf{D}(p \,\|\, r) + \mathsf{D}(q \,\|\, r) = 2\mathsf{D}\Big(\frac{p+q}{2} \,\|\, r\Big) + \mathsf{D}\Big(p \,\|\, \frac{p+q}{2}\Big) + \mathsf{D}\Big(q \,\|\, \frac{p+q}{2}\Big). \tag{E.34}$$

Does the equality hold if we replace the relative entropy by the norm-2 squared? Figure E.4 illustrates a particular example of this identity. Note, in the example we have

$$\begin{aligned}\|p-r\|^2 &= \big\|\big(p - \frac{p+q}{2}\big) + \big(\frac{p+q}{2} - r\big)\big\|^2 \\ &= \big\|p - \frac{p+q}{2}\big\|^2 + \big\|\frac{p+q}{2} - r\big\|^2 - 2\cos(\pi-\theta)\big\|p - \frac{p+q}{2}\big\|\big\|\frac{p+q}{2} - r\big\|\end{aligned}$$

and

$$\begin{aligned}\|q-r\|^2 &= \big\|\big(q - \frac{p+q}{2}\big) + \big(\frac{p+q}{2} - r\big)\big\|^2 \\ &= \big\|q - \frac{p+q}{2}\big\|^2 + \big\|\frac{p+q}{2} - r\big\|^2 - 2\cos(\theta)\big\|q - \frac{p+q}{2}\big\|\big\|\frac{p+q}{2} - r\big\|\,.\end{aligned}$$

Summing these two quantities shows the identity holds for the example.